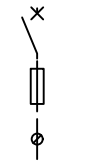
PAGES INDEX & MODIFICATION

SHEET	PAGE TITLE	DESCRIPTION	MODIFICATION									
			V1	V2	V3	V4	V5	V6	V7	V8	V9	V10
1	PAGE INDEX	Page index & modifications	•									
2	LEGEND	Symbols legend, designations	•									
3	POWER DISTRIBUTION	Main incomer Q0, busbars, feeders Q1-Q3	•									
4	BURNER MOTOR POWER	Q1 -> X3 -> P-02 -> RS100 fan motor 3~	•									
5	230VAC CONTROL	BR01 control 230V (Q2, S0, TS)	•									
6	BURNER CABLE P-01	RS100:X1 -> JB-01 -> X2 (cable P-01)	•									
7	BURNER FEEDBACK RELAYS	RR, Rflt 230VAC coils, H3 fault lamp	•									
8	BR02 CLIENT BURNER I/F	BR02 DLX 53: R6/R7 contacts, RR2/Rflt2/Rfuel2, H4, X6	•									
9	24VDC SUPPLY	PSU G1, Q4, fuses FU1-FU5, 0V-PE bond	•									
10	DIGITAL INPUTS - CPU	CPU 1214C DI I0.0-I1.5 -> XI	•									
11	DIGITAL INPUTS - SM1221	SM1221 DI I12.0-I12.7 -> XI2	•									
12	DIGITAL OUTPUTS	CPU 1214C DQ Q0.0-Q1.1 -> R1-R10 -> XQ	•									
13	ANALOG INPUTS	SM 1231 TC 8 x thermocouple -> X5	•									
14	PANEL GA - EXTERNAL	Door front view and side view 1:6, enclosure 600x800x300	•									
15	PANEL GA - INTERNAL	Mounting plate layout and gland plate 1:5	•									
16	TERMINAL SCHEDULE	X2, X3, X4, X5, X6, XI, XI2, XQ	•									
17			•									
18												
19												

POWER HOUSE
POWER WITHOUT LIMITS

PROJECT: Burner PetroBaker	SHEET NAME: PAGE INDEX	PREPARED BY: Eng. Omar Hamed	DATE: 13/10/2026	SHEET: 1 / 16
DRAWING No.: PH-LCP-01		APPROVED BY: Eng. Ahmed Zain		REV: V1

SYMBOLS



Circuit breaker (MCB), 1 pole shown - Q0-Q4

Fuse / fuse terminal - FU

Terminal, strip:number - X2:3, XI:7

Junction (connected) - crossing without dot = not connected



Relay / contactor coil - R, RR, KM

NO contact (normally open)

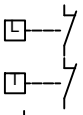
NC contact (normally closed)

Emergency stop, mushroom, latching (NC) - S0

Push button, momentary (NO) - S1, S2

Push button, momentary (NC) - S3

Pressure switch - B1 PSH



Level switch (float) - B2 LSH, B3 LSHH

Thermostat / temperature switch - TS

Signal lamp - H

Solenoid valve coil - YV (24VDC)



Motor, three-phase - MV



Power supply AC/DC - G1

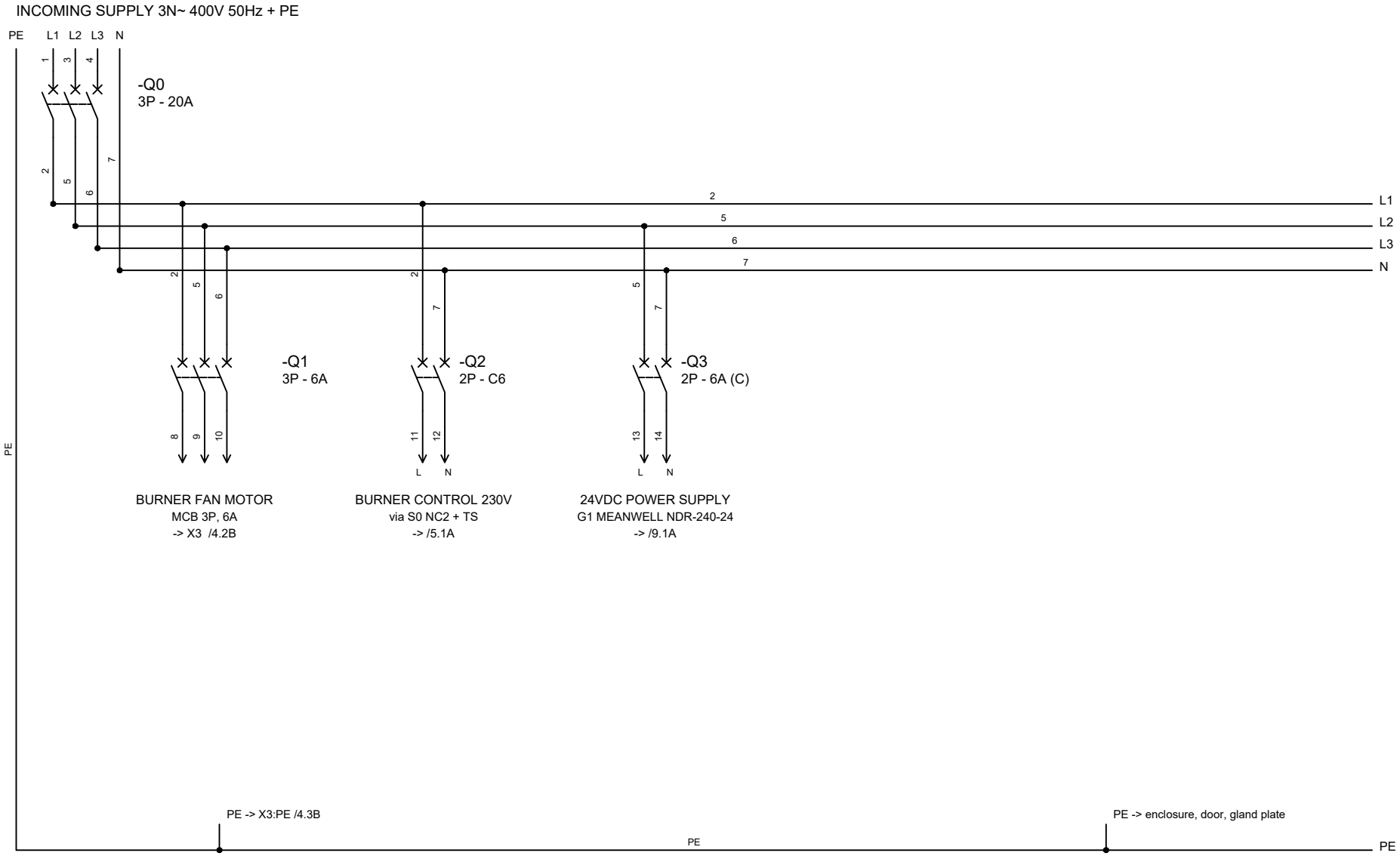
Thermocouple - TE / TT

Protective earth / 0V-PE bond

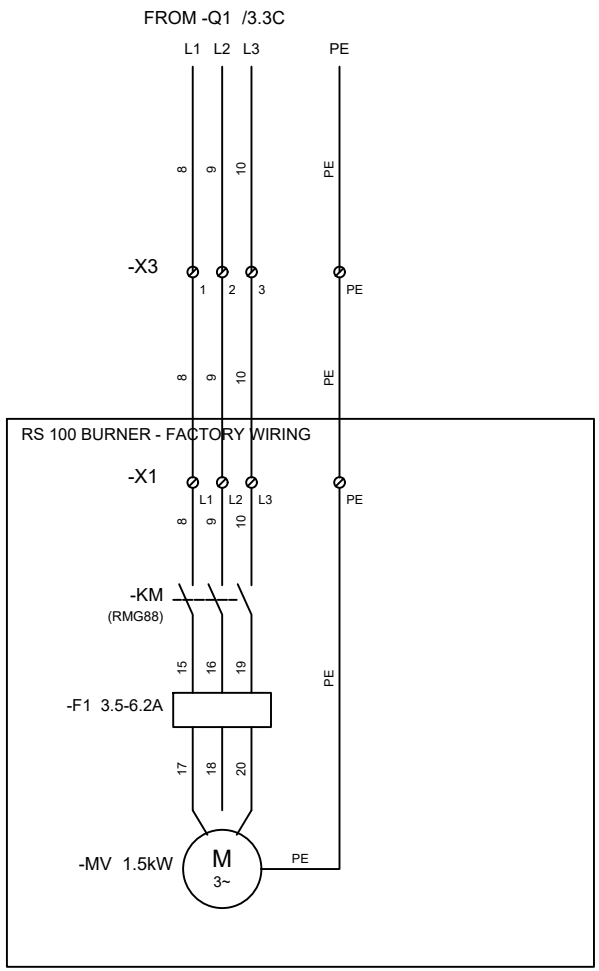
Cable designation - P-01, C-DO-01

Mechanical link / field boundary (dashed)

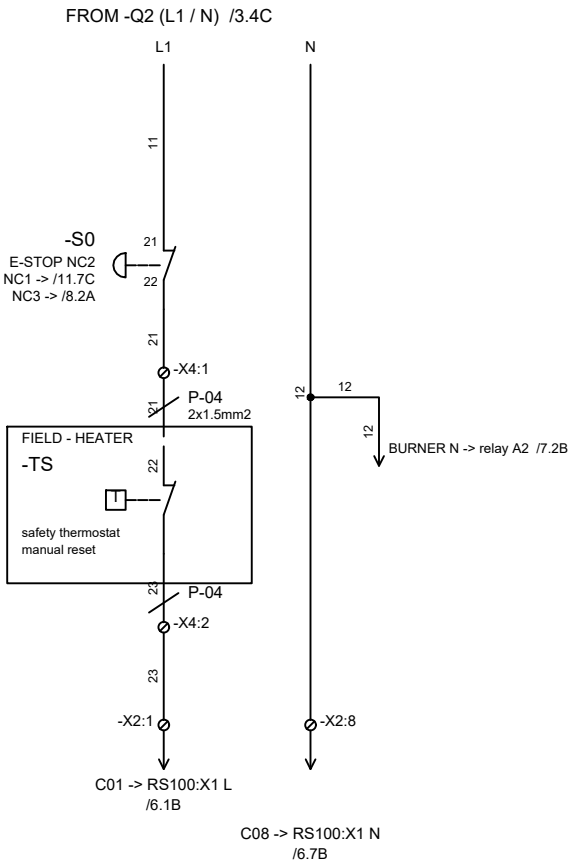
POWER HOUSE — Power Without Limits	PROJECT: Burner PetroBaker	SHEET NAME: Legends	PREPARED BY: Eng. Omar Hamed	DATE: 13/10/2026	SHEET: 2 / 16
	DRAWING No.: PH-LCP-01		APPROVED BY: Eng. Ahmed Zain		REV: V1



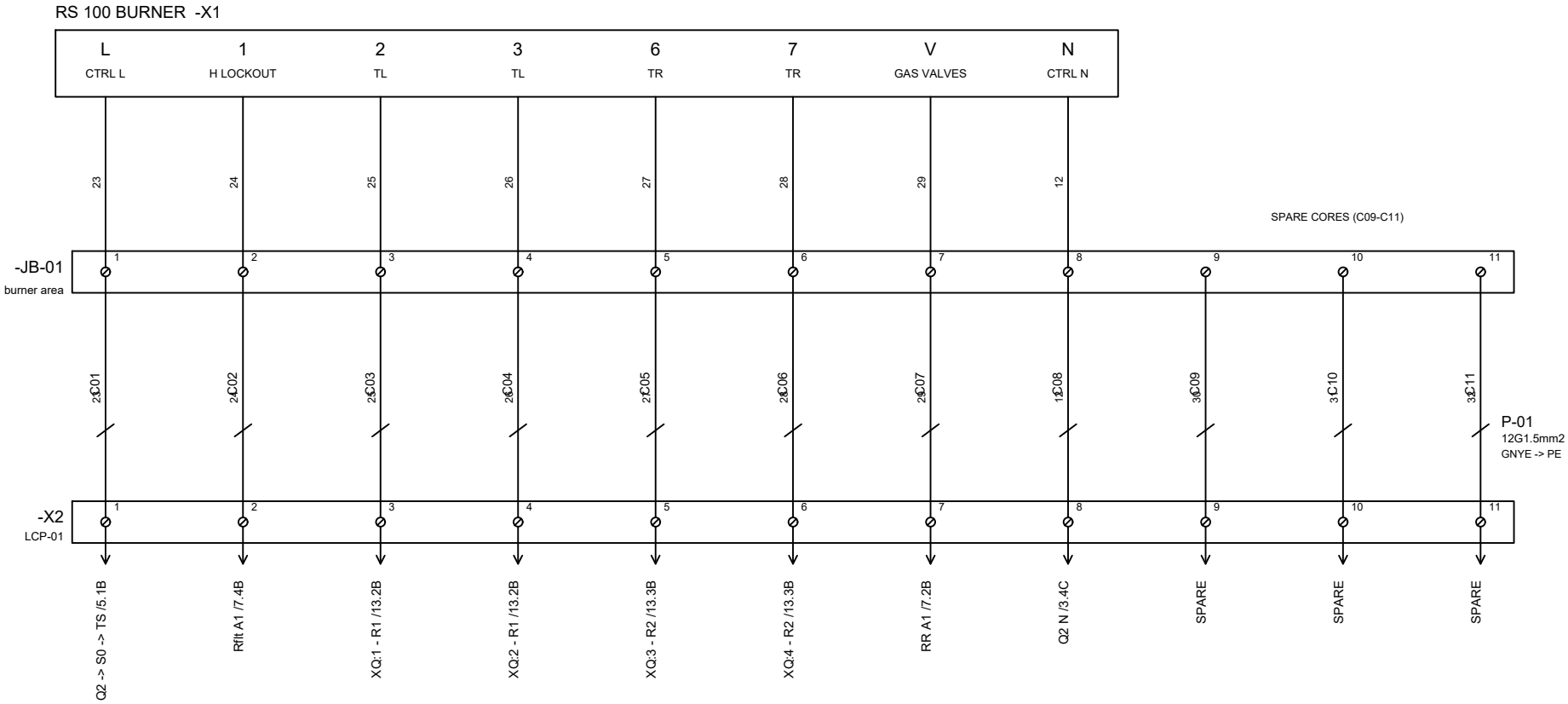
POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME POWER DISTRIBUTION	PREPARED BY Eng. Omar Hamed	DATE 13/10/2026	SHEET 3 / 16
	DRAWING No. PH-LCP-01		APPROVED BY Eng. Ahmed Zain		REV V1



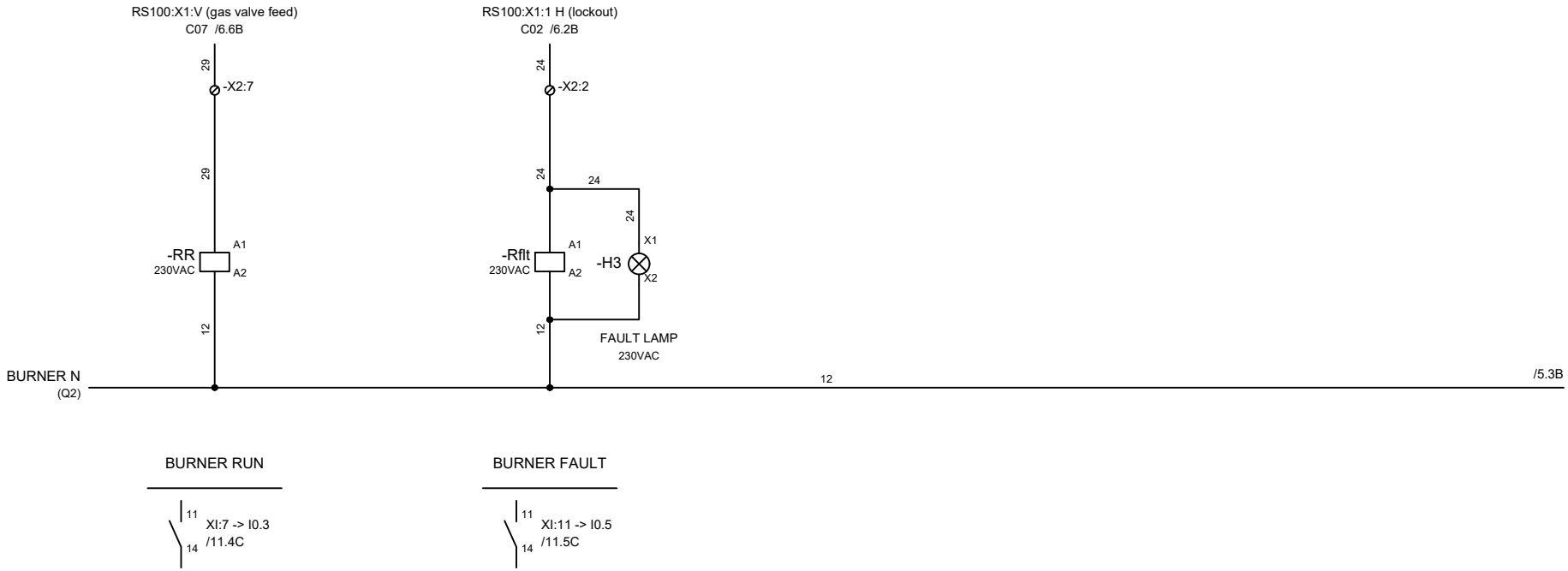
POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME BURNER MOTOR POWER	PREPARED BY Eng. Omar Hamed	DATE 13/10/2026	SHEET 4 / 16
	DRAWING No. PH-LCP-01		APPROVED BY Eng. Ahmed Zain		REV V1



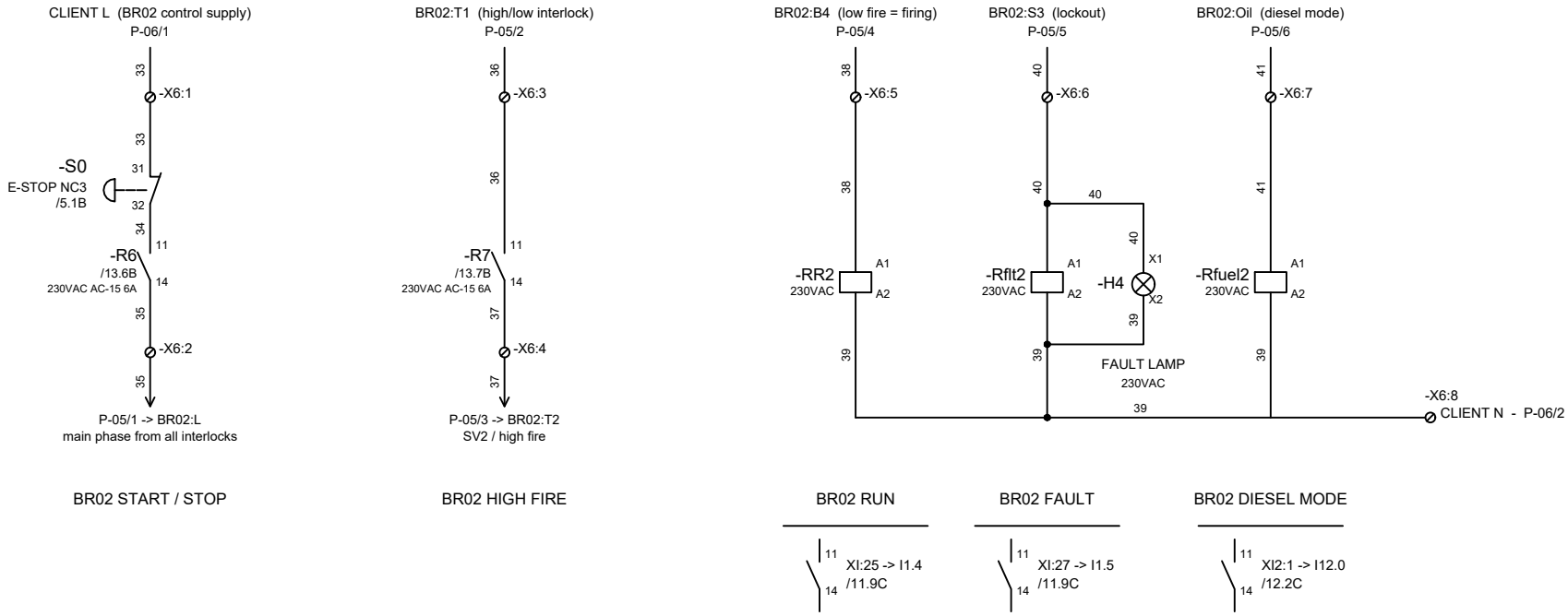
POWER HOUSE — Power Without Limits	PROJECT: Burner PetroBaker	SHEET NAME: AC CONTROL	PREPARED BY: Eng. Omar Hamed	DATE: 13/10/2026	SHEET: 5 / 16
	DRAWING No.: PH-LCP-01		APPROVED BY: Eng. Ahmed Zain		REV: V1



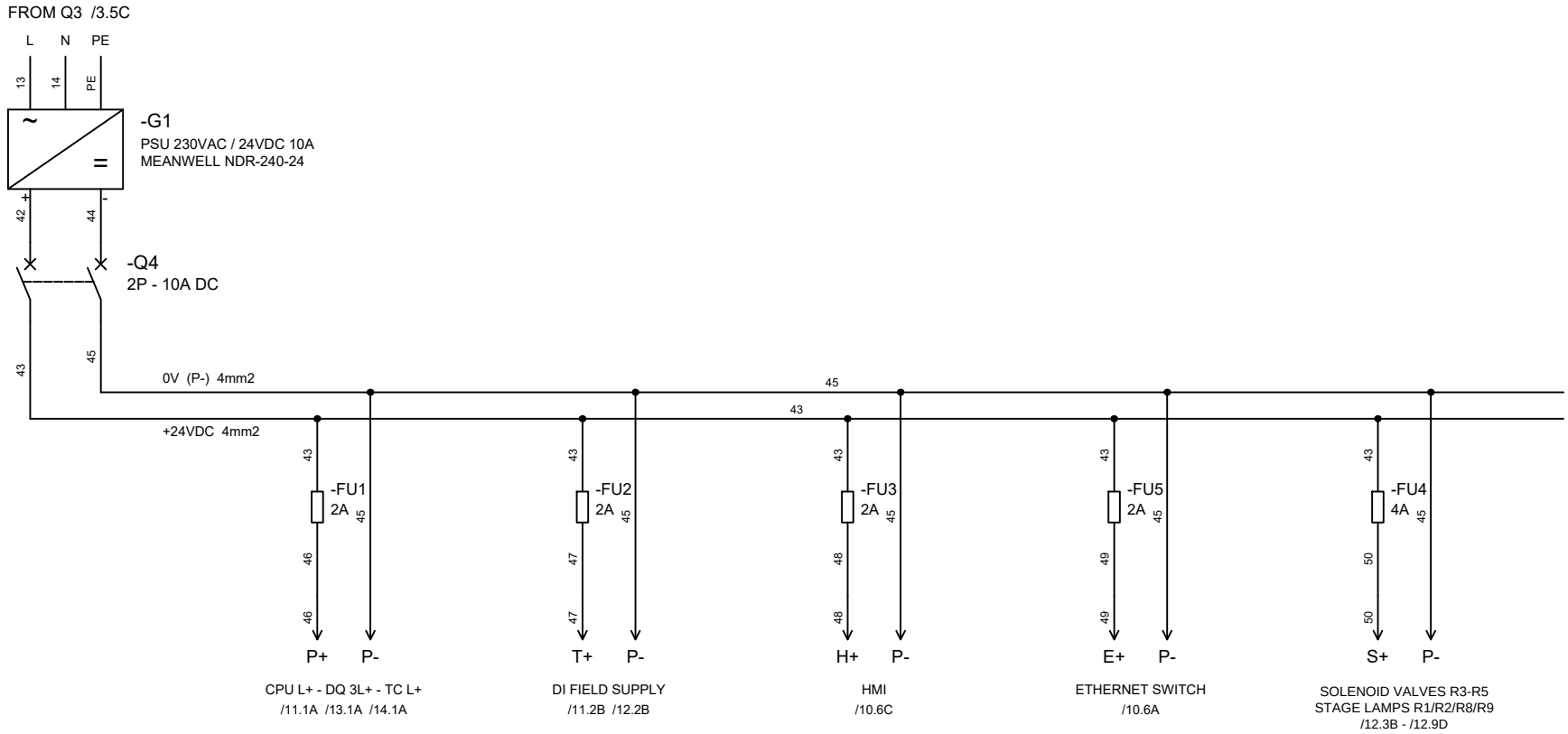
POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME	PREPARED BY Eng. Omar Hamed	DATE	SHEET 6 / 16
	DRAWING No. PH-LCP-01	BURNER CONTROL	APPROVED BY Eng. Ahmed Zain	13/10/2026	REV V1



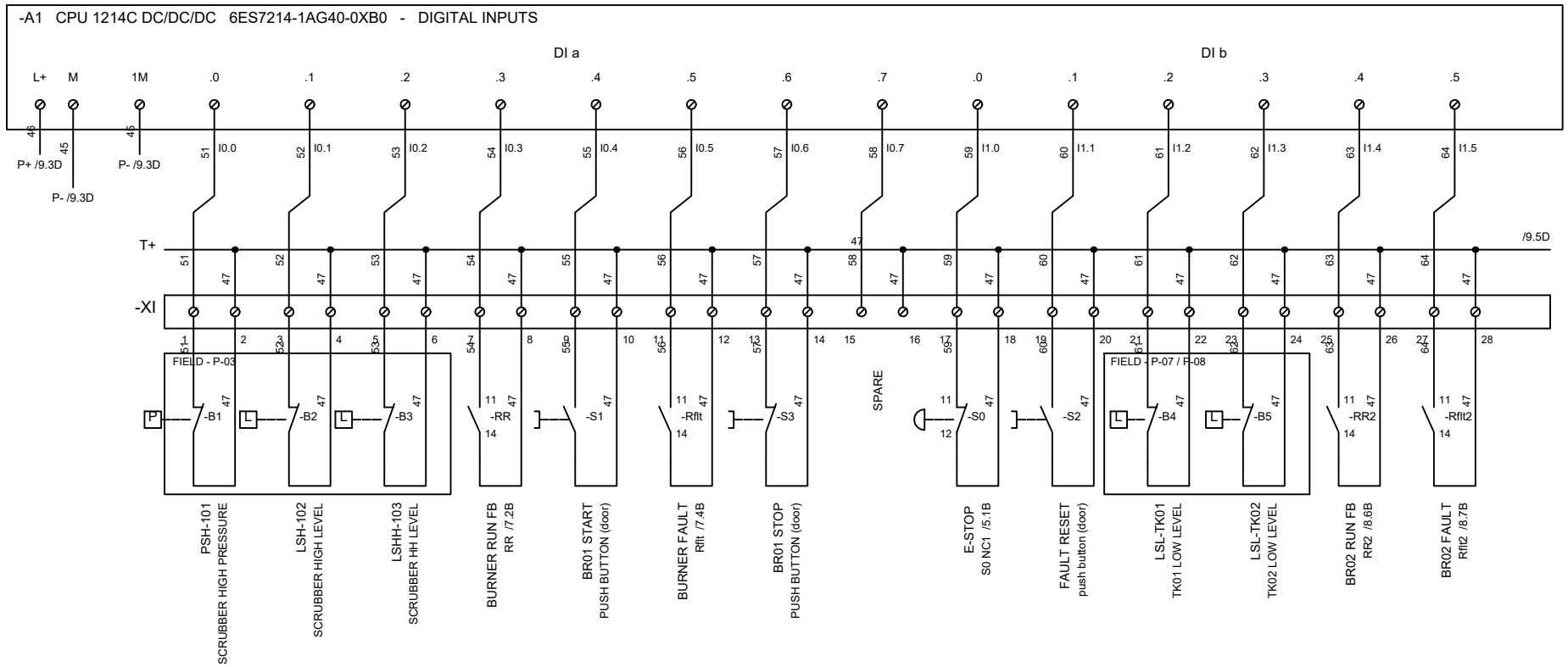
POWER HOUSE — Power Without Limits	PROJECT: Burner PetroBaker	SHEET NAME: BURNER FEEDBACK	PREPARED BY: Eng. Omar Hamed	DATE: 13/10/2026	SHEET: 7 / 16
	DRAWING No.: PH-LCP-01		APPROVED BY: Eng. Ahmed Zain		REV: V1



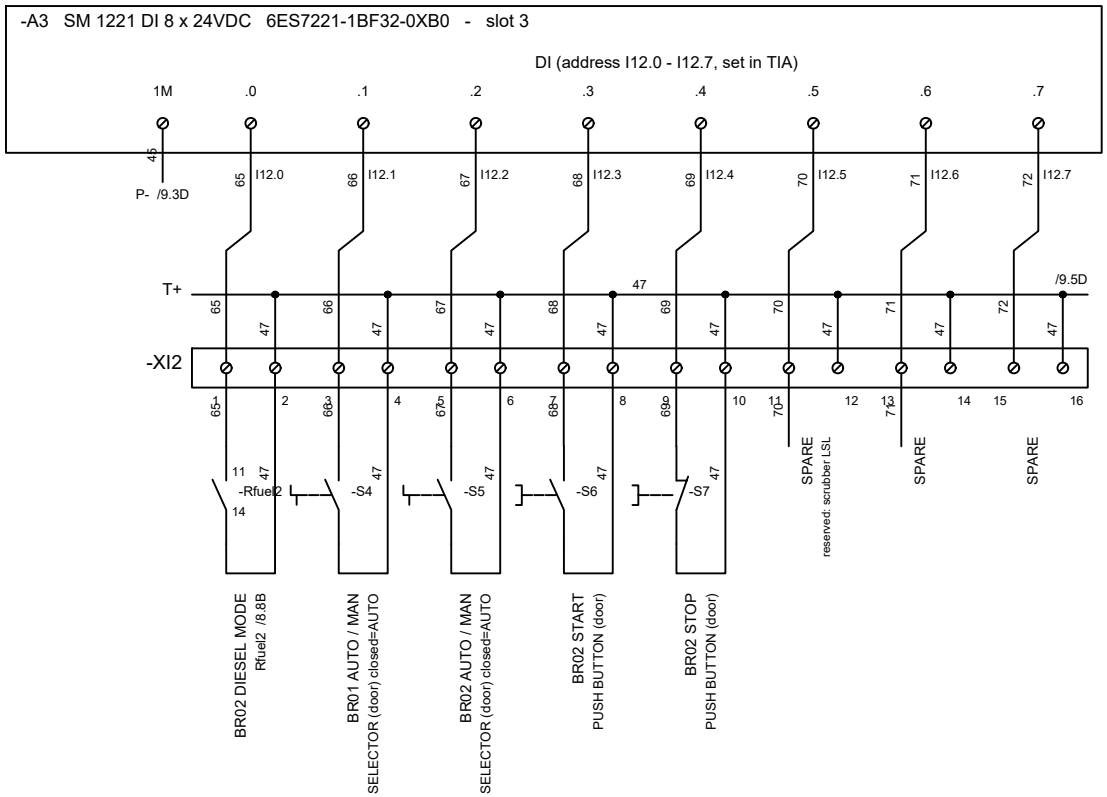
POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME	PREPARED BY Eng. Omar Hamed	DATE	SHEET 8 / 16
	DRAWING No. PH-LCP-01	Burner 2	APPROVED BY Eng. Ahmed Zain	13/10/2026	REV V1



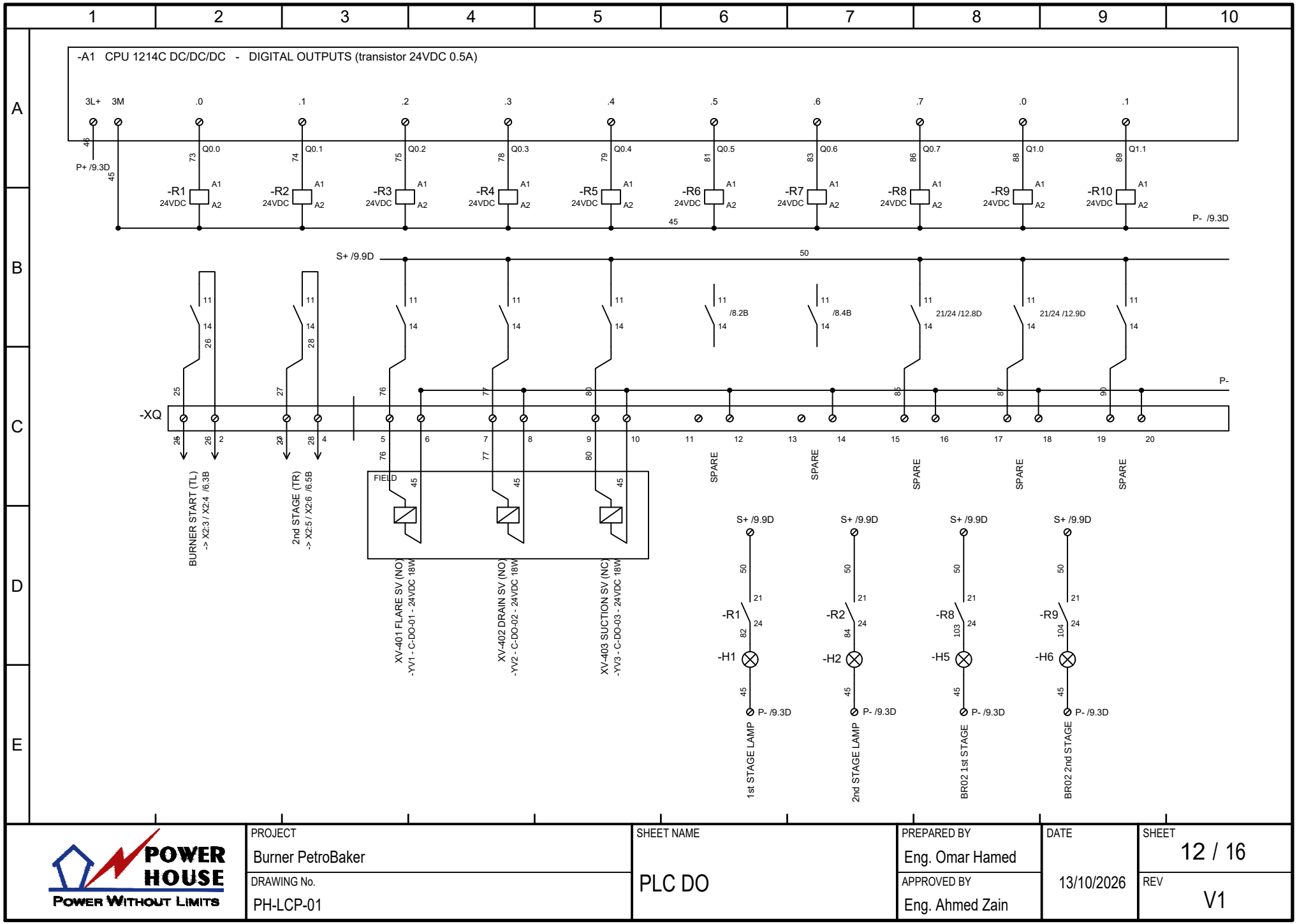
POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME 24VDC Supply	PREPARED BY Eng. Omar Hamed	DATE 13/10/2026	SHEET 9 / 16
	DRAWING No. PH-LCP-01		APPROVED BY Eng. Ahmed Zain		REV V1



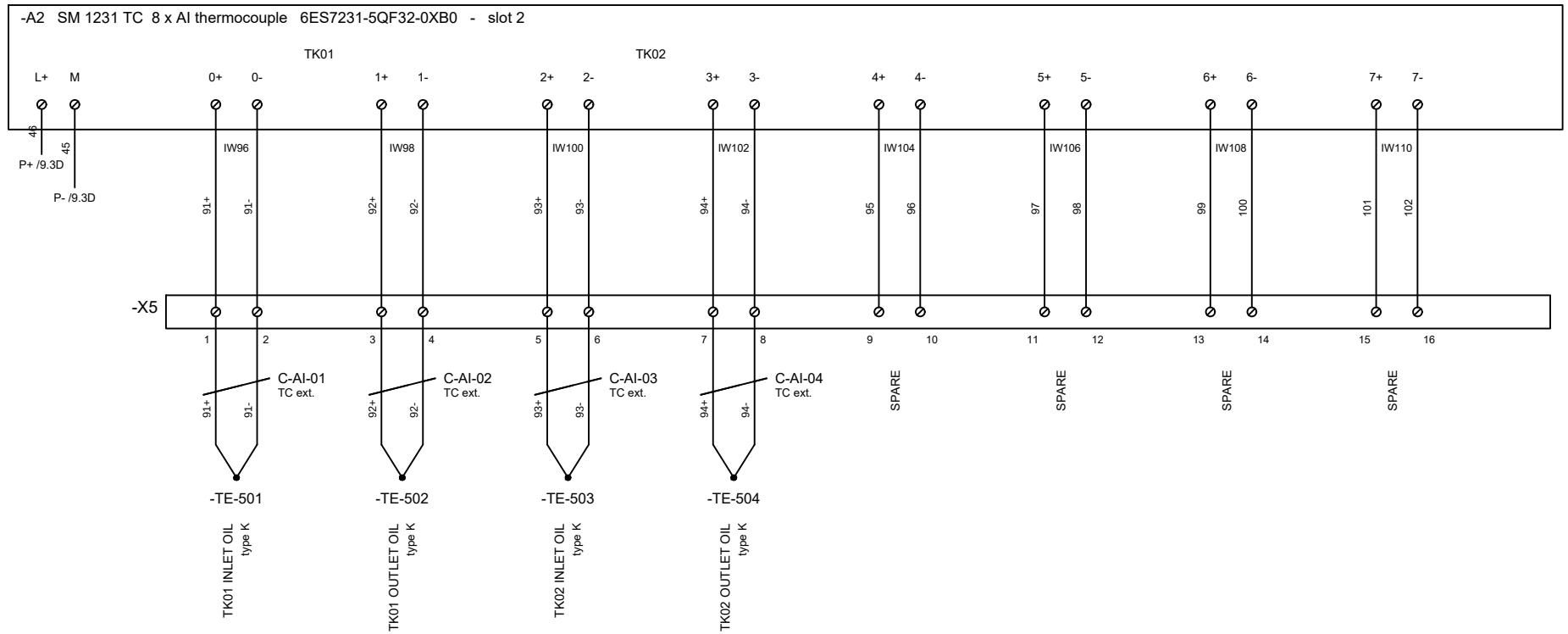
POWER HOUSE — Power Without Limits	PROJECT: Burner PetroBaker	SHEET NAME: PLC DI	PREPARED BY: Eng. Omar Hamed	DATE: 13/10/2026	SHEET: 10 / 16
	DRAWING No.: PH-LCP-01		APPROVED BY: Eng. Ahmed Zain		REV: V1



POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME	PREPARED BY Eng. Omar Hamed	DATE	SHEET 11 / 16
	DRAWING No. PH-LCP-01	DI SM12211	APPROVED BY Eng. Ahmed Zain	13/10/2026	REV V1

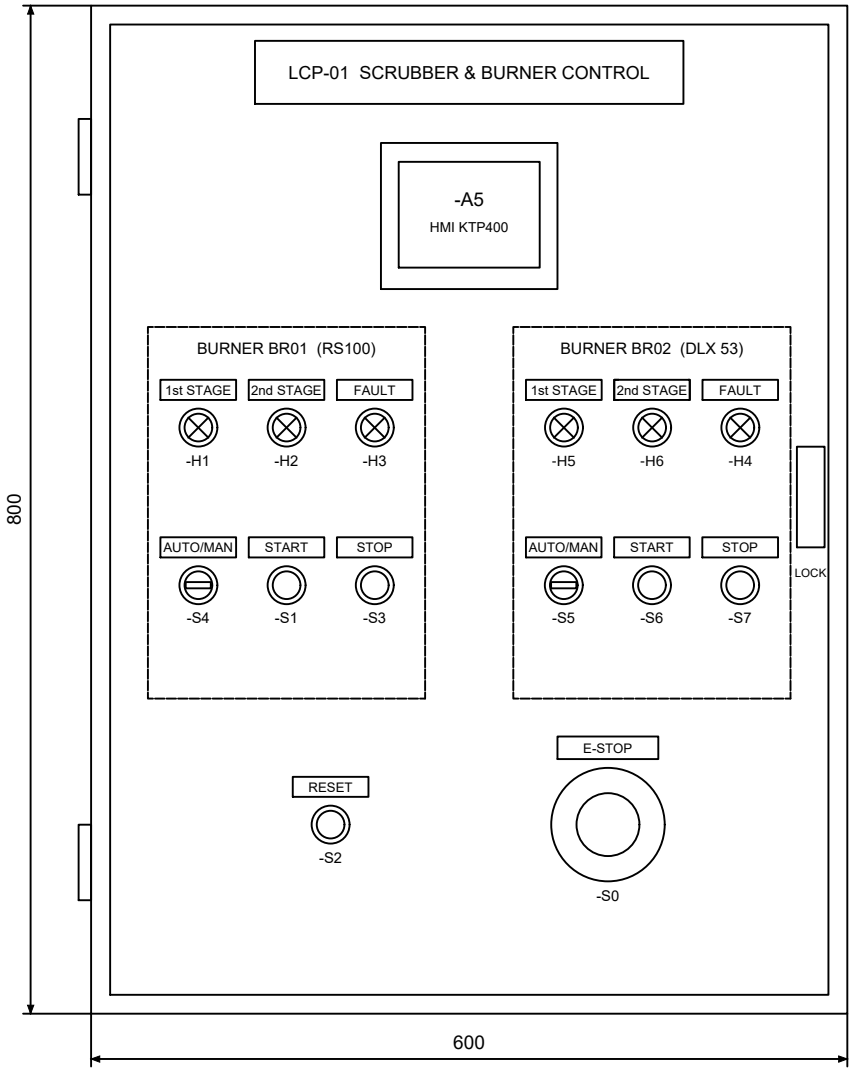


POWER HOUSE — Power Without Limits	PROJECT: Burner PetroBaker	SHEET NAME: PLC DO	PREPARED BY: Eng. Omar Hamed	DATE: 13/10/2026	SHEET: 12 / 16
	DRAWING No.: PH-LCP-01		APPROVED BY: Eng. Ahmed Zain		REV: V1

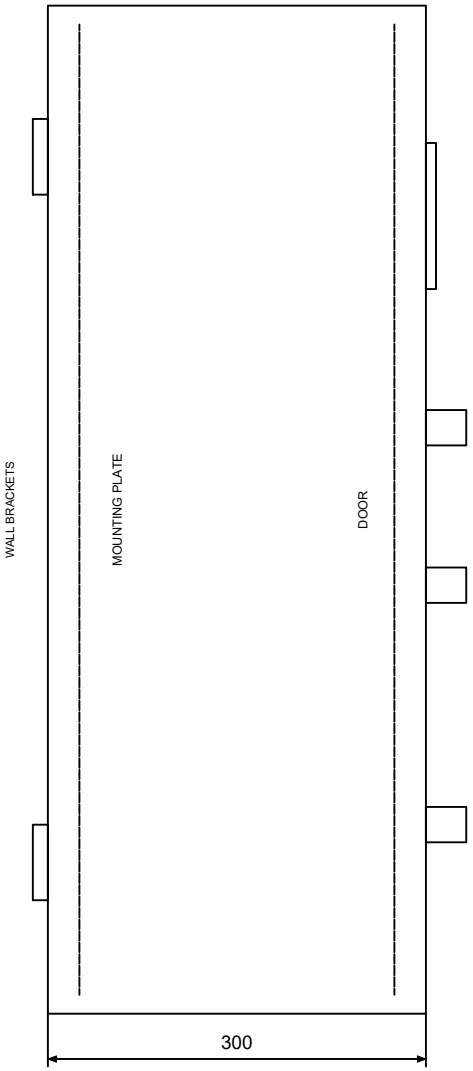


POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME AI TC	PREPARED BY Eng. Omar Hamed	DATE	SHEET 13 / 16
	DRAWING No. PH-LCP-01		APPROVED BY Eng. Ahmed Zain	13/10/2026	REV V1

FRONT VIEW (door closed) 1:6



SIDE VIEW (right) 1:6



ENCLOSURE

600 x 800 x 300 mm

(W x H x D)

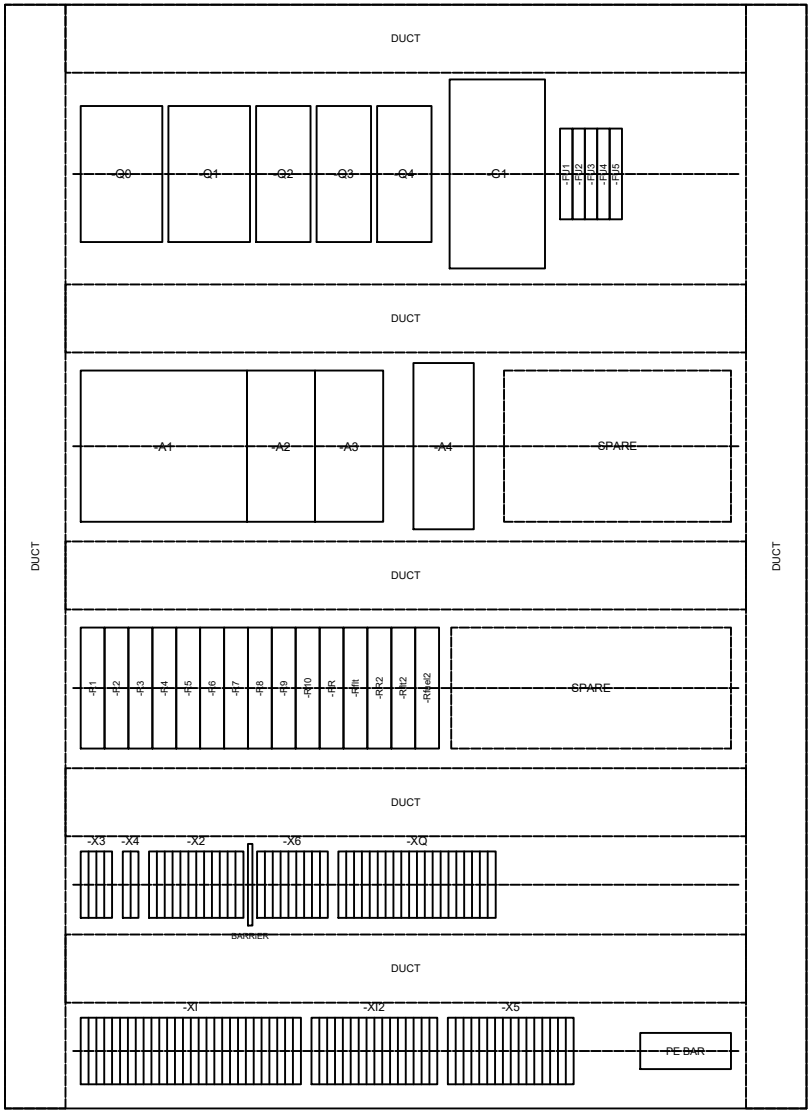
wall-mounted steel

IP65, RAL 7035

device list: /17

POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME	PREPARED BY Eng. Omar Hamed	DATE	SHEET 14 / 16
	DRAWING No. PH-LCP-01	Panel GA	APPROVED BY Eng. Ahmed Zain	13/10/2026	REV V1

MOUNTING PLATE LAYOUT 1:5



POWER HOUSE Power Without Limits	PROJECT Burner PetroBaker	SHEET NAME	PREPARED BY Eng. Omar Hamed	DATE	SHEET 15 / 16
	DRAWING No. PH-LCP-01	Panel Ga Internal	APPROVED BY Eng. Ahmed Zain	13/10/2026	REV V1

-X2 BURNER CONTROL (P-01)

TERM.	INSIDE (panel)	OUTSIDE (field)	WIRE	CABLE
X2:1	Q2 -> S0 -> TS	RS100:X1:L		C01
X2:2	Rflt A1	RS100:X1:1		C02
X2:3	XQ:1 - R1	RS100:X1:2		C03
X2:4	XQ:2 - R1	RS100:X1:3		C04
X2:5	XQ:3 - R2	RS100:X1:6		C05
X2:6	XQ:4 - R2	RS100:X1:7		C06
X2:7	RR A1	RS100:X1:V		C07
X2:8	Q2 N	RS100:X1:N		C08
X2:9	SPARE	JB-01 (spare)		C09
X2:10	SPARE	JB-01 (spare)		C10
X2:11	SPARE	JB-01 (spare)		C11
X2:PE	PE BAR	JB-01 PE		P-01/GNYE

-X3 BURNER MOTOR POWER (P-02)

TERM.	INSIDE (panel)	OUTSIDE (field)	WIRE	CABLE
X3:1	Q1 L1	RS100:X1 L1		P-02/1
X3:2	Q1 L2	RS100:X1 L2		P-02/2
X3:3	Q1 L3	RS100:X1 L3		P-02/3
X3:PE	PE BAR	RS100:X1 PE		P-02/PE

-X4 SAFETY THERMOSTAT (P-04)

TERM.	INSIDE (panel)	OUTSIDE (field)	WIRE	CABLE
X4:1	S0:22 (NC2)	TS contact		P-04/1
X4:2	X2:1	TS contact		P-04/2

-X5 THERMOCOUPLES

TERM.	INSIDE (panel)	OUTSIDE (field)	WIRE	CABLE
X5:1/2	SM1231 0+/0-	TE-501 +/- (TK01 inlet)		C-AI-01
X5:3/4	SM1231 1+/1-	TE-502 +/- (TK01 outlet)		C-AI-02
X5:5/6	SM1231 2+/2-	TE-503 +/- (TK02 inlet)		C-AI-03
X5:7/8	SM1231 3+/3-	TE-504 +/- (TK02 outlet)		C-AI-04
X5:9-16	SM1231 4…7	SPARE		

-X6 BR02 CLIENT BURNER - FOREIGN 230VAC

TERM.	INSIDE (panel)	OUTSIDE (field)	WIRE	CABLE
X6:1	S0:31 (NC3)	Client L in		P-06/1
X6:2	R6:14	BR02:L		P-05/1
X6:3	R7:11	BR02:T1		P-05/2
X6:4	R7:14	BR02:T2		P-05/3
X6:5	RR2:A1	BR02:B4		P-05/4
X6:6	Rflt2:A1, H4:X1	BR02:S3		P-05/5
X6:7	Rfuel2:A1	BR02:Oil		P-05/6
X6:8	RR2/Rflt2/Rfuel2:A2	Client N		P-06/2
X6:PE	PE BAR	BR02 E / client PE		P-05/GNYE

-XI PLC DIGITAL INPUTS

TERM.	INSIDE (panel)	OUTSIDE (field)	WIRE	CABLE
XI:1	CPU I0.0	B1 PSH-101		P-03
XI:2	T+	field supply		P-03
XI:3	CPU I0.1	B2 LSH-102		P-03
XI:4	T+	field supply		P-03
XI:5	CPU I0.2	B3 LSHH-103		P-03
XI:6	T+	field supply		P-03
XI:7	CPU I0.3	RR BURNER RUN FB		
XI:8	T+	field supply		
XI:9	CPU I0.4	S1 BURNER START		
XI:10	T+	field supply		
XI:11	CPU I0.5	Rflt BURNER FAULT		
XI:12	T+	field supply		
XI:13	CPU I0.6	S3 BURNER STOP		
XI:14	T+	field supply		
XI:15	CPU I0.7	SPARE		
XI:16	T+			
XI:17	CPU I1.0	S0 E-STOP (status)		
XI:18	T+	field supply		
XI:19	CPU I1.1	S2 FAULT RESET		
XI:20	T+	field supply		
XI:21	CPU I1.2	B4 LSL-TK01		P-07
XI:22	T+	field supply		P-07
XI:23	CPU I1.3	B5 LSL-TK02		P-08
XI:24	T+	field supply		P-08
XI:25	CPU I1.4	RR2 BR02 RUN FB		
XI:26	T+	field supply		
XI:27	CPU I1.5	Rflt2 BR02 FAULT		
XI:28	T+	field supply		

-XQ PLC DIGITAL OUTPUTS

TERM.	INSIDE (panel)	OUTSIDE (field)	WIRE	CABLE
XQ:1	R1:14	X2:3		internal
XQ:2	R1:11	X2:4		internal
XQ:3	R2:14	X2:5		internal
XQ:4	R2:11	X2:6		internal
XQ:5	R3:14 (S+)	XV-401 +		C-DO-01
XQ:6	P-	XV-401 -		C-DO-01
XQ:7	R4:14 (S+)	XV-402 +		C-DO-02
XQ:8	P-	XV-402 -		C-DO-02
XQ:9	R5:14 (S+)	XV-403 +		C-DO-03
XQ:10	P-	XV-403 -		C-DO-03
XQ:11	not used	SPARE		
XQ:12	P-	SPARE		
XQ:13	not used	SPARE		
XQ:14	P-	SPARE		
XQ:15	R8:14 (S+)	SPARE		
XQ:16	P-	SPARE		
XQ:17	R9:14 (S+)	SPARE		
XQ:18	P-	SPARE		
XQ:19	R10:14 (S+)	SPARE		
XQ:20	P-	SPARE		

-XI2 SM1221 DIGITAL INPUTS

TERM.	INSIDE (panel)	OUTSIDE (field)	WIRE	CABLE
XI2:1	SM1221 I12.0	Rfuel2 BR02 DIESEL		
XI2:2	T+	field supply		
XI2:3	SM1221 I12.1	S4 BR01 AUTO/MAN		
XI2:4	T+	field supply		
XI2:5	SM1221 I12.2	S5 BR02 AUTO/MAN		
XI2:6	T+	field supply		
XI2:7	SM1221 I12.3	S6 BR02 START		
XI2:8	T+	field supply		
XI2:9	SM1221 I12.4	S7 BR02 STOP		
XI2:10	T+	field supply		
XI2:11-16	SM1221 I12.5-I12.7	SPARE (I12.5 reserved)		

POWER HOUSE
POWER WITHOUT LIMITS

PROJECT: Burner PetroBaker	SHEET NAME: TERMINALS	PREPARED BY: Eng. Omar Hamed	DATE: 13/10/2026	SHEET: 16 / 16
DRAWING No.: PH-LCP-01		APPROVED BY: Eng. Ahmed Zain		REV: V1